



EXPERIMENT 10 — MATLAB

This is the **last experiment** in the uploaded set. It has three parts: **CNN Image Classification, CNN Image Segmentation, and CNN Object Detection**. MATLAB-exp10

Ex. 1 — CNN for Image Classification

This is the code shown in your PDF.

```
clc;
clear;
close all;

% Load Dataset
[images, digits, angles] = digitTrain4DArrayData;

disp('Dataset Loaded Successfully');

% Load pretrained AlexNet
net = alexnet;

disp('AlexNet Loaded Successfully');

% Display Network
analyzeNetwork(net);
```

What it does

```
digitTrain4DArrayData
  ↓
Loads digit image dataset
  ↓
alexnet
  ↓
Loads pretrained AlexNet
  ↓
analyzeNetwork(net)
  ↓
Displays CNN architecture
```

Your PDF shows:

Dataset Loaded Successfully

AlexNet Loaded Successfully

and then MATLAB's **Deep Learning Network Analyzer**. MATLAB-exp10

Ex. 2 — CNN for Image Segmentation

Copy this directly:

```
clc;  
clear;  
close all;  
  
imageSize = [224 224 3];  
  
numClasses = 2;  
  
net = unet(imageSize,numClasses);  
  
analyzeNetwork(net);
```

What it does

```
imageSize = [224 224 3]
```

means:

```
Height = 224  
Width  = 224  
Channels = 3
```

Then:

```
numClasses = 2;
```

creates a segmentation network with **2 classes**.

Finally:

```
net = unet(imageSize,numClasses);
```

creates the U-Net architecture and:

```
analyzeNetwork(net);
```

opens the network architecture in MATLAB.

This is the code shown in the supplied Exp. 10 PDF. MATLAB-exp10

Ex. 3 — CNN for Object Detection

This is the **object-detection code shown in your PDF**:

```
clc;
clear;
close all;

disp('CNN for Object Detection');

% Check if YOLOv2 detector is available
which yolov2ObjectDetector

% Load pretrained backbone
net = alexnet;

disp('Pretrained AlexNet Loaded Successfully');

% Display network architecture
analyzeNetwork(net);

disp('YOLOv2 Object Detection Toolbox is Available');

disp('Training images and bounding box labels are required to train a detector.');
```

```
disp('Object detection can be performed after training the network.');
```

Expected idea/output

The PDF shows MATLAB checking:

which yolov2ObjectDetector

and then displaying the pretrained AlexNet architecture.

It also states that **training images and bounding-box labels are required to train the detector** before actual object detection can be performed. MATLAB-exp10

 **One important thing for Exp. 10**

These commands require the appropriate **MATLAB Deep Learning / Computer Vision functionality and pretrained-network support**.

For example:

```
alexnet
```

may require the **Deep Learning Toolbox Model for AlexNet Network** to be installed.

And:

```
unet(...)
```

requires the relevant Deep Learning Toolbox functionality.

So if MATLAB gives you something like:

```
Undefined function 'alexnet'
```

or

```
Undefined function 'unet'
```

your code isn't necessarily wrong — the required toolbox/model package isn't installed.